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DATA PREPROCESSING AND EDA OF BELGRADE AIR POLLUTION DATA

By Areeba Ahmed

Project Report On

Data Preprocessing and EDA of Belgrade Air Pollution Data

Data Mining (119161)

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SUMMARY

This report presents a data mining project on the Belgrade air pollution dataset from December 2021 to March 2022. The primary objective is to perform data preprocessing and exploratory data analysis (EDA) using Python tools such as pandas, sklearn.preprocessing, and matplotlib. The dataset includes daily measurements of pollutants like SO_2, PM_{10}, NO_2, NO_x, CO, NO, $PM_{2.5}$, along with weather variables including pressure (P), temperature (T), and relative humidity (RH).

The analysis involved loading the dataset, handling preprocessing tasks such as scaling and missing value checks, and conducting EDA through statistical descriptions, visualizations like histograms, correlation heatmaps, pairplots, and boxplots. Manual calculations were also performed on the first 10 records to understand basic statistical measures.

Findings show strong correlations among pollutants such as NO_2 and NO_x, suggesting common sources like vehicle emissions, while weather variables had weak influence on pollution levels during this specific time period. Some outliers, especially in NO_x and NO, indicate short-term spikes likely caused by traffic or specific events.

Overall, this project demonstrates how proper preprocessing and EDA help in understanding real-world environmental data, providing meaningful insight into Belgrade's air quality. The study also builds a solid base for future work such as prediction or clustering models to support better air quality management and planning.

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1. Introduction

Air pollution is a critical environmental issue affecting urban areas worldwide, with significant consequences for public health, ecosystems, and climate. Exposure to pollutants such as particulate matter ($PM_{2.5}$, PM_{10}), sulfur dioxide (SO_2), nitrogen oxides (NO_x), and carbon monoxide (CO) has been linked to respiratory and cardiovascular diseases, reduced life expectancy, and environmental degradation. Belgrade, the capital of Serbia, experiences fluctuating air quality levels due to a combination of industrial emissions, traffic density, and seasonal weather patterns that can trap pollutants near the ground. Monitoring and analyzing air quality in this context is essential for understanding pollution dynamics and supporting data-driven policy decisions to improve urban air quality.

This project focuses on a dataset containing daily air quality measurements from Belgrade spanning late 2021 to early 2022. The dataset includes 81 records with variables such as SO_2 (sulfur dioxide), PM_{10} (particulate matter of 10 micrometers), NO_2 (nitrogen dioxide), NO_x (nitrogen oxides), CO (carbon monoxide), NO (nitric oxide), $PM_{2.5}$ (particulate matter of 2.5 micrometers), atmospheric pressure (P), temperature (T), and relative humidity (RH). The primary aim of this report is to perform data preprocessing and exploratory data analysis (EDA) as part of a data mining study. Preprocessing involves cleaning, transforming, and preparing the dataset to ensure accuracy and consistency, while EDA helps uncover patterns, trends, anomalies, and relationships among the measured variables.

Python will be used as the primary tool for analysis, with libraries such as pandas for data manipulation, `sklearn.preprocessing` for scaling and normalization, and `matplotlib/seaborn` for visualization. Through techniques such as histograms, boxplots, correlation heatmaps, and pairplots, the study will investigate the distribution of pollutants, identify potential outliers, and examine correlations between air pollutants parameters and meteorological conditions. The insights gained from this analysis aim to provide a structured understanding of Belgrade's air pollution patterns and support further research, public health assessment, and environmental policy development.

2. Methods

This section describes the step-by-step procedures used for data preprocessing and exploratory data analysis (EDA) on the Belgrade air pollution dataset. All steps were performed using Python.

2.1. Data Loading

The dataset `"azzsdata2022.txt"` was loaded into a pandas DataFrame. It contains daily air quality measurements for Belgrade, including pollutants such as SO_2, PM10, NO_2, NOx, CO, NO, PM2.5, as well as meteorological variables like temperature, pressure, and relative humidity.

2.2. Inspect Dataset Structure

To understand the dataset before preprocessing, the DataFrame's information was examined using the `info()` method. The dataset contains 81 records and 12 features, all corresponding to Belgrade. The `city` column contains a constant value across all records. No missing values were detected in any of the columns.

2.3. Drop Unnecessary Columns

Since the `city` column contains the same value for all records, it does not provide any analytical value and was dropped from the dataset. The `time` column, being a temporal index rather than a quantitative measurement, was retained as a reference for potential temporal analysis but excluded from numerical statistical calculations. As a result, all descriptive statistics and visualizations were computed using only the remaining numerical features.

2.4. Handling of Missing Values

The dataset was examined for missing or null values using `df.info()`, and no missing values were found in any column. Therefore, no imputation was required.

2.5. Generation of Descriptive Statistics

Summary statistics were computed for all numerical features using the `df.describe()` method. This provided measures of central tendency and dispersion, including mean, median, standard deviation, minimum, maximum, and quartiles.

2.6. Data Type Conversion

The `time` column was converted to datetime format to make it easier to create plots, like the monthly distribution of records.

2.7. Exploratory Data Analysis Visualizations

A comprehensive suite of visualizations was created to explore the dataset's structure and relationships. The distribution of all numerical variables was examined through histograms generated with `plt.hist()` providing insight into their underlying frequency distributions and skewness. To quantify and visualize inter-variable relationships, a correlation matrix was computed and displayed as a heatmap using `sns.heatmap()` highlighting significant positive and negative correlations between pollutants and meteorological factors. Multivariate patterns and potential clustering were investigated using `sns.pairplot()`, which created a matrix of scatter plots for pairwise feature analysis. Finally, boxplots produced were utilized for robust outlier detection and to summarize the central tendency and dispersion of each variable, clearly illustrating medians, quartiles, and potential anomalous data points.

2.8. Feature Scaling

To complete preprocessing, all numerical features were standardized using `StandardScaler` from `sklearn.preprocessing` (Z-score). Since the features are in different units and ranges and some algorithms are sensitive to scale, standardization brings all features to a comparable range, giving each feature a mean of 0 and a standard deviation of 1. The scaled dataset was prepared for further analysis, while all visualizations and descriptive statistics were computed on the original data.

2.9. Manual Calculation of First 10 Records

A manual data description and exploratory data analysis was performed on the first ten records of the dataset. Calculated mean, standard deviation, minimum, maximum, quartiles (Q1, Q2, Q3), interquartile range (IQR), and lower and upper fences for outlier detection for all numerical variables. Additionally, basic visualizations such as histograms and boxplots were constructed manually to develop an intuitive understanding of the data distributions and to identify potential outliers.

3. Discussion

This section presents the evidence from the analysis, including data inspections, preprocessing results, and EDA insights. All findings are based on the executed Python code.

3.1. Initial Data Inspection and Cleaning

An initial examination of the Belgrade air pollution dataset revealed several important characteristics about its structure and content. . The dataset consists of 81 daily records covering multiple pollutants (SO_2, PM_{10}, NO_2, NO_x, CO, NO, $PM_{2.5}$) and environmental variables (temperature, pressure, relative humidity). Pollutant concentrations are measured in μg/m³ (except CO in mg/m³), temperature in °C, pressure in hPa, and relative humidity in %. A preview of the first few records via `df.head()` confirmed the proper formatting and allowed an initial assessment of values.

	city	time	so2	pm10	no2	nox	co	no	pm25	p	t	rh
0	Belgrade	30/12/2021	7.35	36.92	45.49	127.47	0.88	53.37	32.32	1010.22	6.20	90.07
1	Belgrade	31/12/2021	20.87	89.90	70.12	414.02	2.04	224.31	77.31	1012.67	8.79	82.54
2	Belgrade	01/01/2022	10.93	64.43	38.83	133.12	1.30	61.54	59.86	1012.87	10.04	82.36
3	Belgrade	02/01/2022	15.21	42.57	55.74	198.03	1.12	92.96	37.21	1012.73	11.14	75.14
4	Belgrade	03/01/2022	13.99	68.64	63.15	292.04	1.54	149.28	60.49	1006.57	9.09	77.98

Figure 3.1: First Five Records of the Original Dataset

Analysis of the dataset structure using `df.info()` showed that all columns were complete, with no missing or null values. This indicates that the dataset is reliable for subsequent analysis without the need for imputation or data cleaning to handle missing information. Such completeness is beneficial as it ensures that computed statistics and visualizations are based on the full set of observations.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 81 entries, 0 to 80
Data columns (total 12 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   city    81 non-null     object
 1   time    81 non-null     object
 2   so2     81 non-null     float64
 3   pm10    81 non-null     float64
 4   no2     81 non-null     float64
 5   nox     81 non-null     float64
 6   co      81 non-null     float64
 7   no      81 non-null     float64
 8   pm25    81 non-null     float64
 9   p       81 non-null     float64
 10  t       81 non-null     float64
 11  rh      81 non-null     float64
dtypes: float64(10), object(2)
memory usage: 7.7+ KB
```

Figure 3.2: Dataset Structure and Data Types

During inspection, it was found that the city column had the same value, “Belgrade” in every record. This shows that all data comes from one location and there is no variation based on city. Because this column does not add any useful information for analysis, it was removed to make the dataset simpler and avoid unnecessary repetition.

```
df.drop(columns=['city'], inplace=True)
```

	time	so2	pm10	no2	nox	co	no	pm25	p	t	rh
0	30/12/2021	7.35	36.92	45.49	127.47	0.88	53.37	32.32	1010.22	6.20	90.07
1	31/12/2021	20.87	89.90	70.12	414.02	2.04	224.31	77.31	1012.67	8.79	82.54
2	01/01/2022	10.93	64.43	38.83	133.12	1.30	61.54	59.86	1012.87	10.04	82.36
3	02/01/2022	15.21	42.57	55.74	198.03	1.12	92.96	37.21	1012.73	11.14	75.14
4	03/01/2022	13.99	68.64	63.15	292.04	1.54	149.28	60.49	1006.57	9.09	77.98

Figure 3.3: Dataset After Removing City Code Column

The *time* column was also examined and converted to a datetime format, enabling accurate temporal analysis. This transformation was necessary for evaluating patterns over time, such as monthly distributions of records or seasonal trends in pollution levels. The conversion ensures that subsequent analyses that rely on chronological ordering, grouping, or plotting are reliable and interpretable.

```
df['time'] = pd.to_datetime(df['time'], dayfirst=True)
```

	time	so2	pm10	no2	nox	co	no	pm25	p	t	rh
0	2021-12-30	7.35	36.92	45.49	127.47	0.88	53.37	32.32	1010.22	6.20	90.07
1	2021-12-31	20.87	89.90	70.12	414.02	2.04	224.31	77.31	1012.67	8.79	82.54
2	2022-01-01	10.93	64.43	38.83	133.12	1.30	61.54	59.86	1012.87	10.04	82.36
3	2022-01-02	15.21	42.57	55.74	198.03	1.12	92.96	37.21	1012.73	11.14	75.14
4	2022-01-03	13.99	68.64	63.15	292.04	1.54	149.28	60.49	1006.57	9.09	77.98

Figure 3.4: Time Column After Conversion to Datetime Format

Overall, this initial inspection and cleaning phase demonstrates that the dataset is structurally sound, complete, and ready for exploratory analysis.

3.2. Temporal Distribution of Records

The monthly distribution of records reveals that the dataset covers a limited time span from December 2021 to March 2022. The highest number of observations was recorded in January and February 2022, while December 2021 contains the fewest entries. This imbalance indicates that the dataset is temporally skewed toward early 2022.

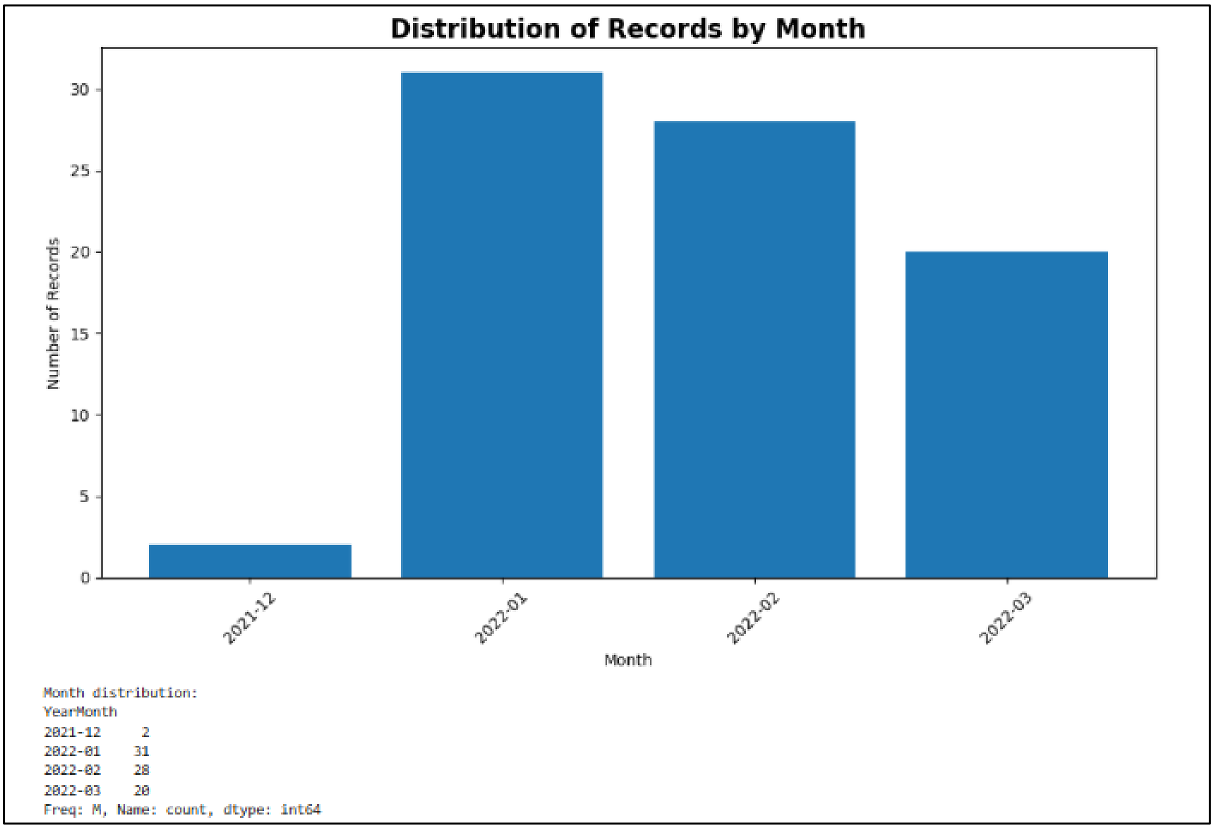

Figure 3.5: Monthly Distribution of Air Pollution Records

3.3. Distribution of Pollutants and Environmental Variables

3.3.1. Visual Distribution Analysis

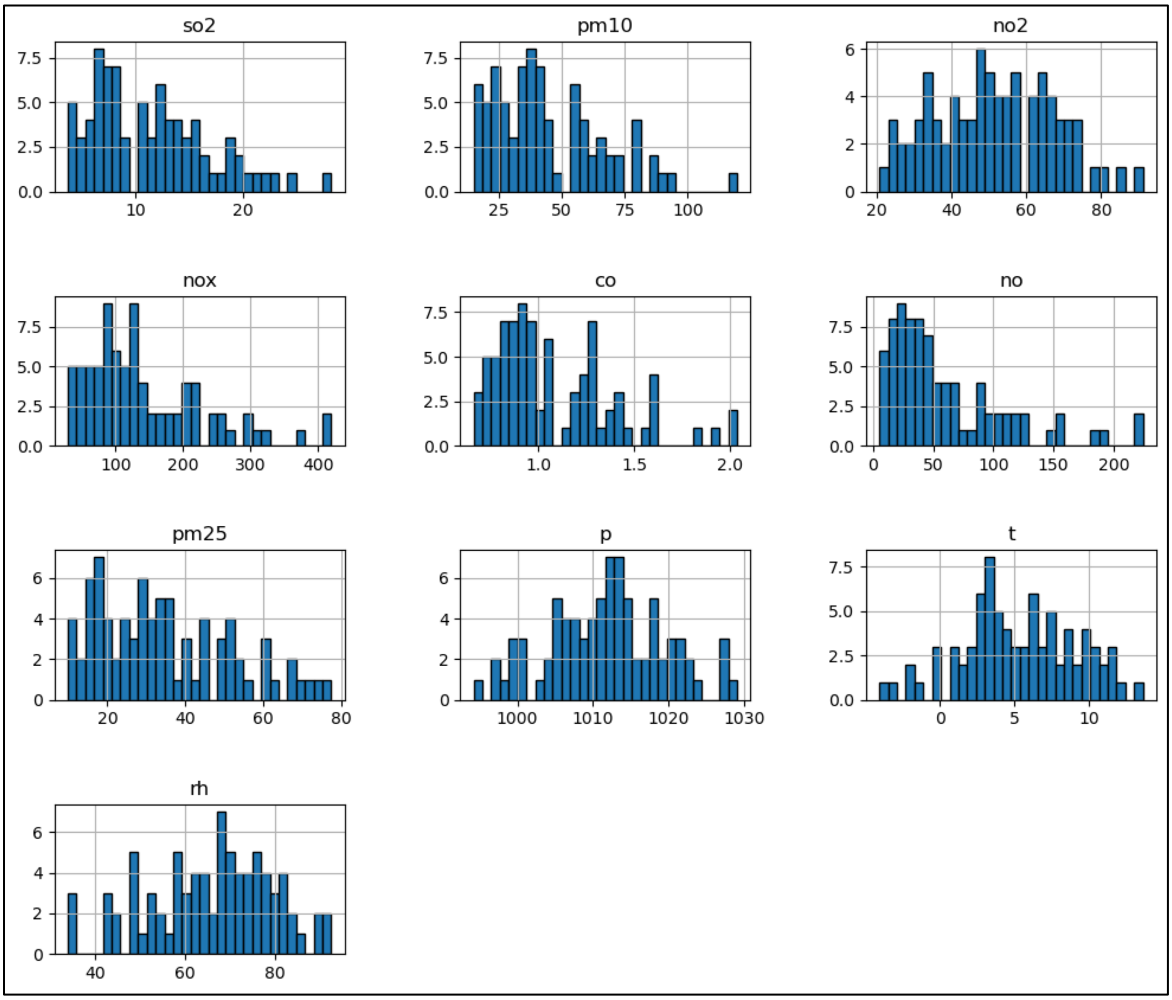

Figure 3.6: Histograms of Air Pollutants and Environmental Variables

The histograms show that most air pollutants in Belgrade exhibit right-skewed distributions, with a few extreme high values. For instance, SO_2 and CO mostly remain low (below 15 µg/m³ and 1.4 mg/m³ respectively), but occasional industrial or heating episodes produce spikes. Similarly, PM_{10} and $PM_{2.5}$ cluster in moderate ranges (15–60 µg/m³ and 10–50 µg/m³) but show occasional peaks above 100 µg/m³ for PM_{10} and 70 µg/m³ for $PM_{2.5}$, indicating frequent moderate pollution with rare severe events. Nitrogen compounds (NO, NO_2, NOx) follow a right-skewed pattern.

In contrast, environmental variables behave differently: temperature (T) shows a right-skewed distribution reflecting mostly cold winter days with occasional warmer days, while relative humidity (RH) is left-skewed with a peak around 70–90%.Atmospheric pressure (P) is roughly symmetric and unimodal, centered near 1010–1020 hPa, with small variability. Overall, the distributions indicate that urban pollutants frequently stay within moderate ranges but have occasional high-impact events, whereas meteorological variables are more stable.

3.3.2. Descriptive Statistical Summary

	so2	pm10	no2	nox	co	no	pm25	p	t	rh
count	81.000000	81.000000	81.000000	81.000000	81.000000	81.000000	81.000000	81.000000	81.000000	81.000000
mean	11.414568	44.628272	51.168642	142.340988	1.082346	60.542716	34.574568	1011.722099	5.257531	65.856049
std	5.456263	21.899804	15.892835	88.143907	0.319563	49.750838	17.209154	7.643771	3.835367	13.967888
min	3.670000	15.020000	20.800000	29.010000	0.660000	5.360000	9.860000	994.210000	-4.070000	33.910000
25%	7.130000	27.840000	38.830000	81.540000	0.850000	26.320000	19.990000	1006.560000	3.000000	57.510000
50%	10.930000	39.120000	50.920000	121.790000	0.960000	43.180000	31.760000	1012.320000	5.240000	67.840000
75%	15.120000	57.950000	63.210000	198.030000	1.270000	86.080000	45.580000	1016.860000	8.180000	76.340000
max	28.260000	119.560000	91.140000	419.510000	2.040000	224.310000	77.310000	1029.090000	13.610000	92.600000

Figure 3. 7: Data Description of Numerical Variables

The descriptive statistics provide a numerical overview of the central tendency and variability of all pollutants and meteorological variables. All attributes contain 81 observations, indicating complete and consistent records. Among the pollutants, NO_x shows the highest mean concentration (142.34 µg/m³) along with a very large standard deviation (88.14), highlighting strong fluctuations in nitrogen oxide levels, likely influenced by varying traffic intensity and combustion activities. PM_{10} and $PM_{2.5}$ have mean values of 44.62 µg/m³ and 34.57 µg/m³ respectively, suggesting generally moderate particulate pollution, but their relatively high standard deviations indicate occasional pollution peaks, consistent with urban emission events. Similarly, NO and NO_2 display noticeable variability, confirming unstable traffic-related emissions.

In contrast, CO exhibits a comparatively low mean (1.08 mg/m³) and smaller standard deviation, indicating more stable concentration levels with fewer extreme fluctuations. SO_2 also remains relatively low on average (11.41 µg/m³), though the difference between its maximum and third quartile suggests occasional short-term emission spikes. Regarding meteorological variables, atmospheric pressure shows minimal variation (mean ≈ 1011.7 hPa, std ≈ 7.64), highlighting its stability throughout the study period. Temperature and relative humidity demonstrate moderate variability, reflecting natural seasonal and daily weather changes.

Overall, the noticeable gap between maximum values and the upper quartiles for several pollutants confirms the presence of episodic high-pollution events, while the central values indicate that most observations remain within moderate ranges. These statistical patterns reinforce the visual insights obtained from the histograms and boxplots.

3.4. Correlation Analysis

3.4.1. Correlation Heatmap: Strength and Interpretation

The correlation heatmap below illustrates the strength and direction of relationships between pollutants and meteorological variables in the Belgrade air pollution dataset.

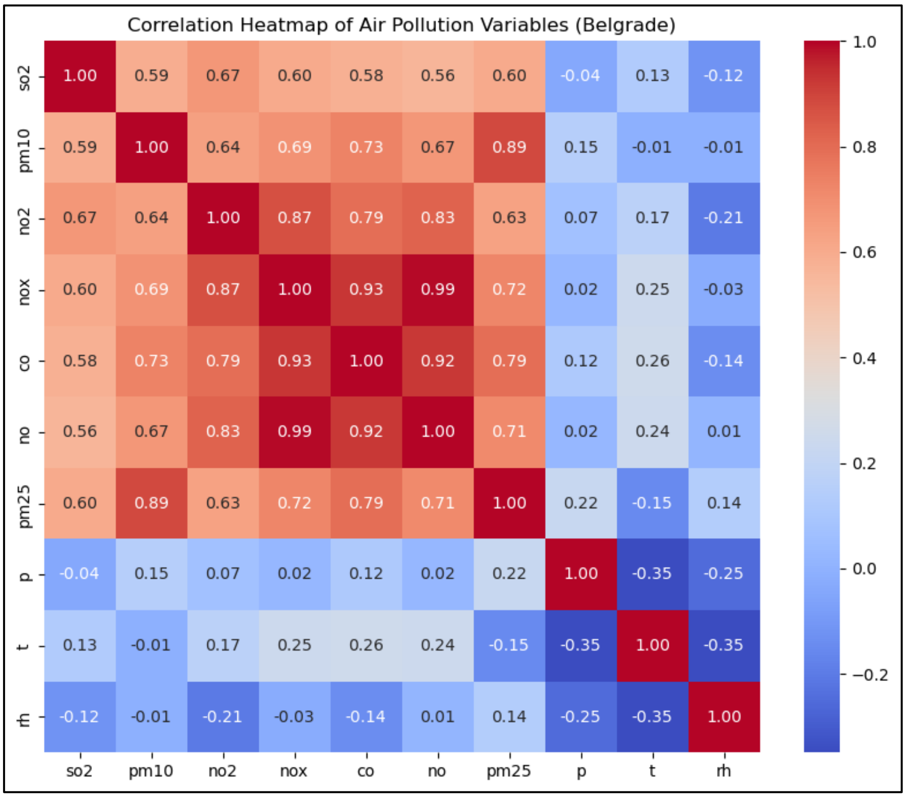

Figure 3.8: Correlation Heatmap (Red = positive, Blue = negative)

- **Combustion-related pollutants**

  As shown in Figure 3.8, NO_x, NO, and CO have very strong positive correlations, with values close to 1. This means when one of these pollutants increases, the others also increase at the same time. This usually happens because they come from the same sources, mainly vehicles and fuel burning for heating. So, higher traffic or more heating activity leads to higher levels of all three gases together.

- **Particulate matter behavior (PM_{10} and $PM_{2.5}$)**

  These particles are strongly correlated with each other and with combustion gases like NOx and CO. This suggests that emissions not only produce gases but also contribute to dust and fine particles in the air.

- **SO_2 behavior and mixed sources**

  SO_2 shows moderate correlations with other pollutants. This indicates that while some SO_2 comes from traffic, it also comes from other sources like factories and power plants. Therefore, its pattern is not as closely linked with the main pollution group.

- **Influence of meteorological variables**

  Temperature, humidity, and air pressure show weak correlations with pollutant. This means weather conditions have a limited effect on pollution compared to human activities.

- **Overall interpretation**

  The heatmap shows that air pollution during the studied winter months is mainly driven by combustion processes, with different pollutants rising together due to common sources. Weather conditions had only a minor influence.

3.4.2. Pairplot: Visual Examination of Variable Relationships

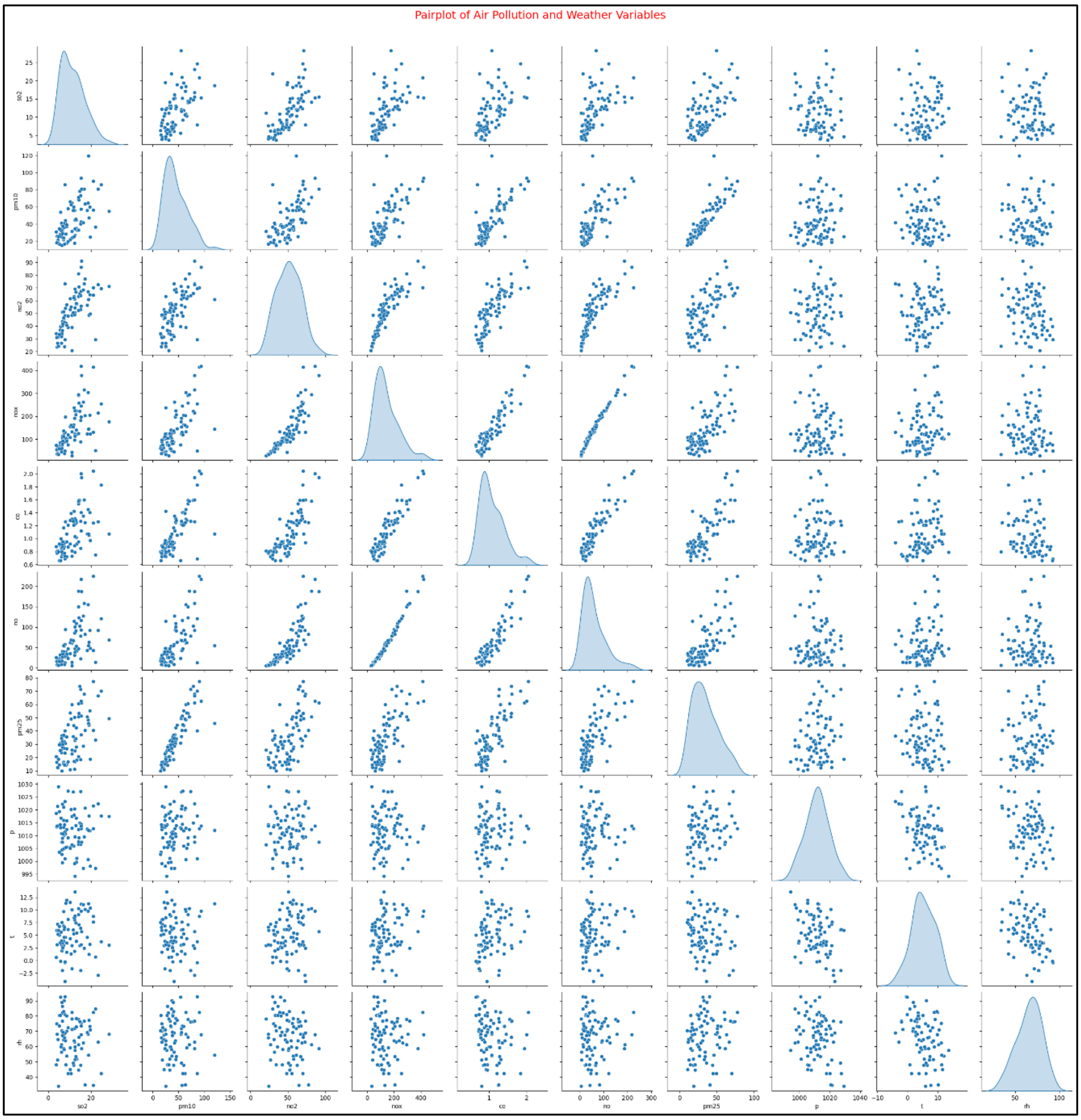

Figure 3.9: Pairplot

The pairplot provides a visual representation of the relationships among pollutants and meteorological variables through scatter plots and distribution histograms along the diagonal. Strong linear patterns were observed between NO_x, NO, and CO, confirming their strong association identified in the heatmap. PM_{10} and $PM_{2.5}$ also displayed dense, positively sloped clusters, visually reinforcing their strong correlation.

In contrast, plots involving temperature, humidity, and pressure show widely scattered point distributions with most pollutants, indicating weak influence on pollution levels. These visual patterns validate that pollution variation is mainly driven by emission sources rather than meteorological conditions during the study period.

3.4.3. Summary of Relationships

- **Very strong positive (>0.90)**: NO_x–NO (0.99), NO_x–CO (0.93), NO–CO (0.92).These indicate a common dominant shared source.
- **Strong positive (0.70–0.89)**: PM_{10}–$PM_{2.5}$ (0.89), NO_2 with NOx, NO and CO, and particulate matter with CO and NOx. This means that particulate matter (PM_{10} and $PM_{2.5}$) mainly comes from combustion sources such as vehicle exhaust and heating systems, as their levels rise and fall together with gases produced during burning.
- **Moderate positive (0.60–0.69)**: SO_2 with most pollutants. This reflects partial shared sources but with weaker consistency.
- **Weak or near-zero correlation (<0.40)**: Meteorological variables with pollutants. This shows that emission intensity dominated over weather effects during this period.

3.5. Boxplots Analysis

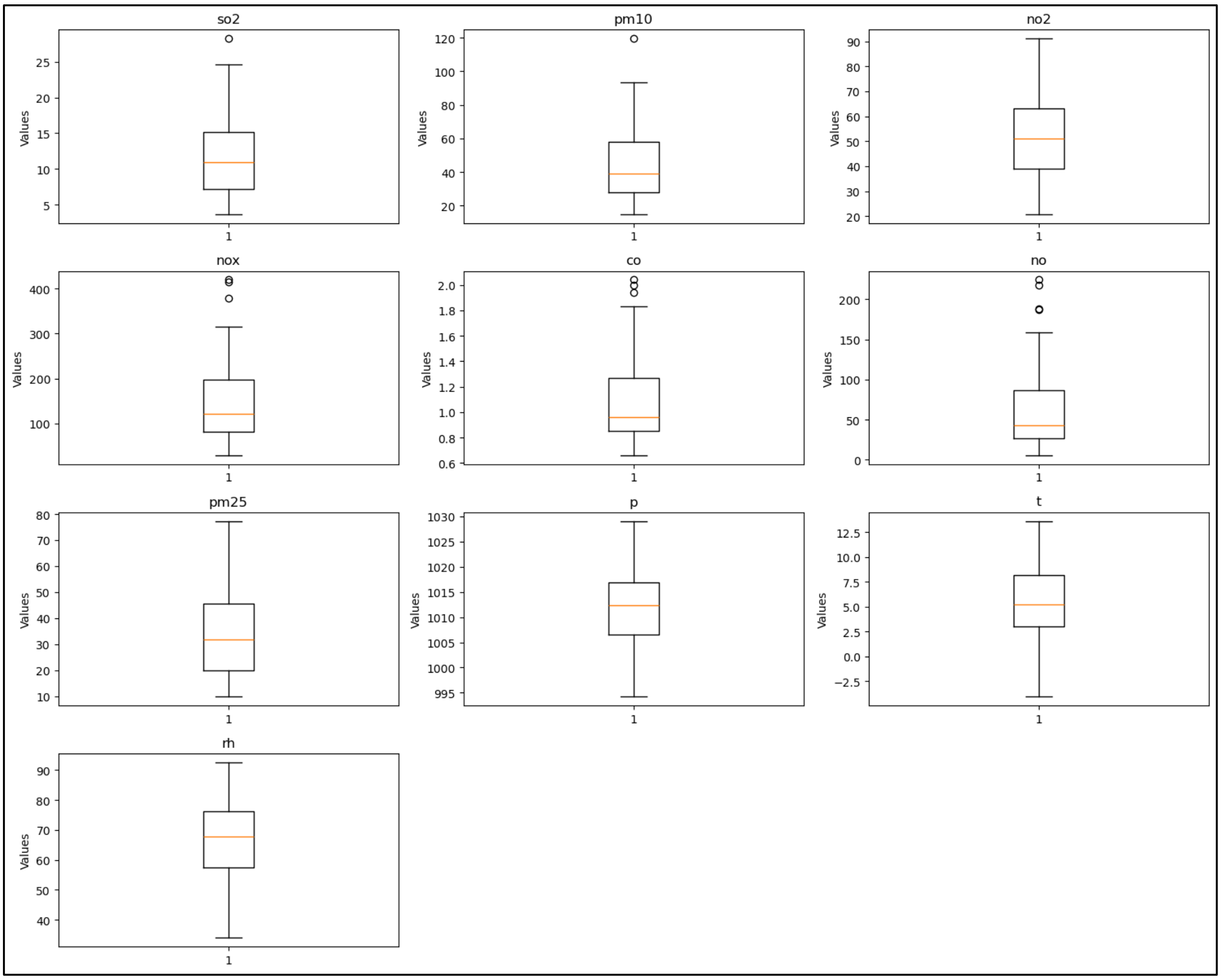

Figure 3.10: Boxplots of all numeric variables

The boxplots of all numerical features provide insights into the distribution, variability, and presence of extreme values in the Belgrade air pollution dataset. They confirm that most observed variations are realistic and correspond to environmental or seasonal phenomena rather than errors or faulty measurements.

- **SO_2 (Sulfur Dioxide)**

  Most values of SO_2 lie within a low and stable range, indicating generally consistent air quality regarding sulfur dioxide. One outlier is present, which likely represents a short-term emission event, such as increased industrial activity or fuel burning during colder days. This outlier reflects real environmental variability rather than a measurement error.

- **PM_{10} (Particulate Matter)**
  The boxplot for PM_{10} shows moderate typical values with a few extreme high points. These spikes are realistic and often occur due to events like dust storms, heavy traffic, or dry weather conditions. Such peaks represent event-based fluctuations rather than data anomalies.

- **NO_2 (Nitrogen Dioxide)**
  NO_2 values appear stable with no significant outliers detected. This suggests consistent air quality regarding nitrogen dioxide during the dataset period.

- **NOx (Nitrogen Oxides)**
  NOx shows strong variability with several extreme values. These outliers are expected because NOx levels are highly sensitive to combustion processes, such as industrial emissions or sudden traffic surges. They reflect genuine short-term environmental changes.

- **CO (Carbon Monoxide)**
  CO values are mostly stable but include a few higher points. These may occur due to incomplete fuel combustion, dense traffic, or heating emissions. Again, these extreme values are realistic observations rather than errors.

- **NO (Nitric Oxide)**
  NO displays a very wide range with some very high values. This gas is highly reactive and fluctuates quickly depending on traffic density, reflecting real-time environmental variability.

- **$PM_{2.5}$ (Fine Particulate Matter)**
  $PM_{2.5}$ exhibits moderate variation with noticeable peaks. Such high values often occur under smog conditions or poor air circulation, particularly during winter months.

- **Temperature (T)**

  Temperature shows some extreme low and high values corresponding to seasonal effects rather than errors. These extremes represent very cold or unusually warm days.

- **Pressure (P)**

  Atmospheric pressure is tightly clustered with almost no extreme points, indicating high stability and consistent measurement throughout the dataset.

- **Relative Humidity (RH)**

  RH demonstrates moderate spread with a few extreme values on both ends. These extremes reflect very dry or very humid days, which are natural seasonal variations.

- **Overall Interpretation**

  The boxplots confirm that the dataset reflects realistic environmental and meteorological behavior. Extreme values generally correspond to actual events or seasonal changes, and there is no evidence of random noise or faulty measurements. Meteorological variables are generally more stable, while pollutants show variability linked to emissions, traffic, or environmental events.

3.6. Scaling

After applying `StandardScaler`, all numerical features were standardized to have mean 0 and standard deviation 1.

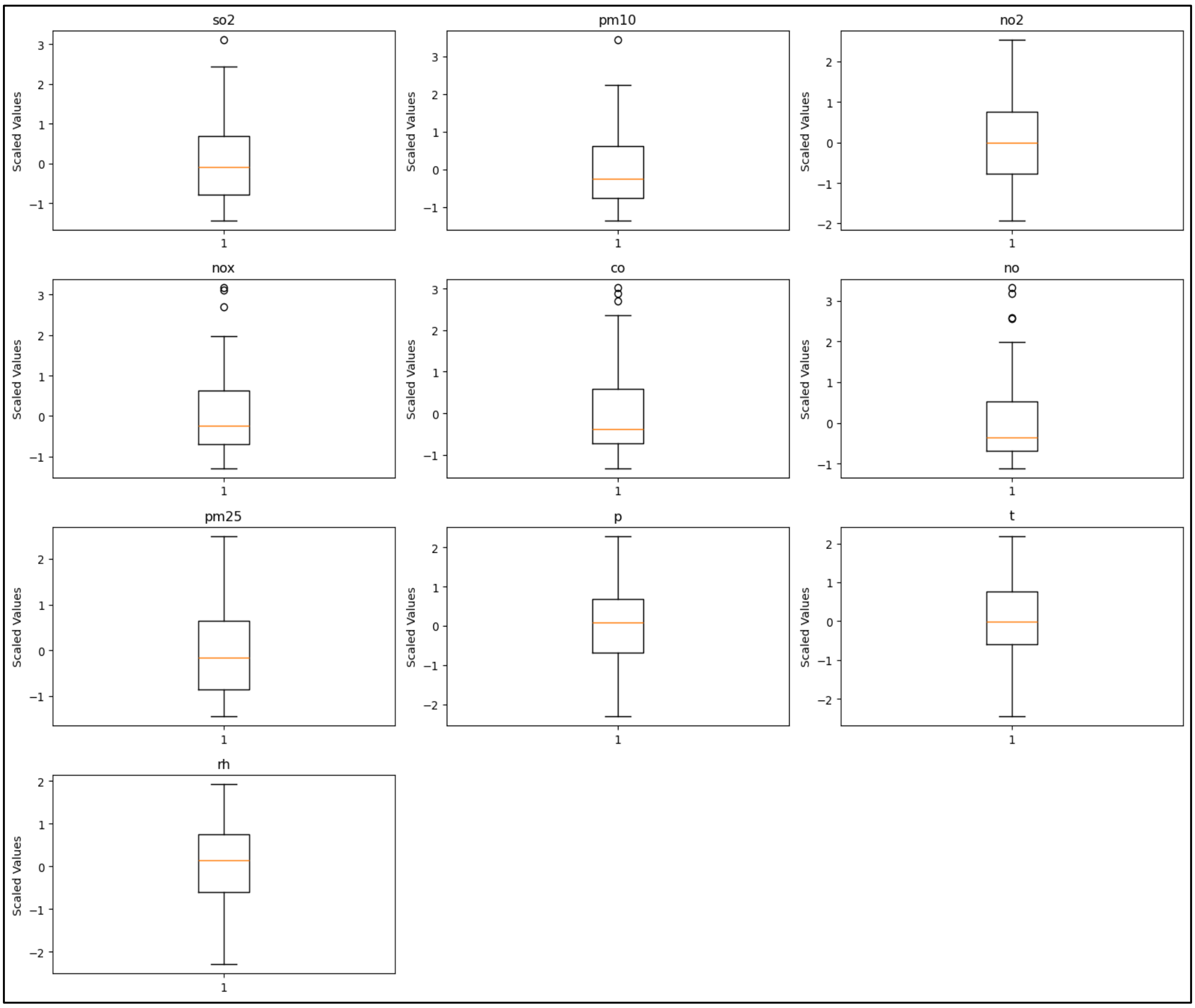

Figure 3. 11: Boxplots of all numerical features after scaling

The boxplot above shows that all features now lie on a similar scale while maintaining their original distribution patterns. This confirms that scaling was applied uniformly across the dataset.

3.7. Comparison of Manual vs Python-based Analysis

When only 10 records were used for manual analysis, single high or low values strongly affected the mean, standard deviation, quartiles, and the appearance of histograms and boxplots, making the results unstable and sometimes misleading. Using the full 81-record dataset gave more reliable and consistent results, with averages, spreads, and quartiles better representing the true behaviour of the data. Although general trends were similar in both cases, such as the strong relationship between NOx and NO, the small dataset showed irregular variations, while the full dataset displayed smoother and more realistic patterns. Relationships between variables were clearer in the larger dataset, and extreme values had less impact as they were balanced by more data points.

Remarkably, even with only the first 10 daily records, the manual boxplots already revealed that traffic-related primary pollutants (NO_x, NO, CO) exhibited wider spreads and longer upper whiskers than meteorological variables and SO_2, correctly indicating their episodic, emission-driven nature. When the full 81-record dataset was analysed using Python, these patterns were confirmed and strengthened, with numerous high outliers appearing in SO_2, NO_x, NO, CO and PM_{10} that were not present in the first 10 days. Meanwhile, meteorological variables maintained nearly identical, compact, and symmetric distributions in both analyses, confirming their stable behaviour throughout the study period. This comparison highlights that manual analysis on small samples is useful for learning, but larger datasets are needed for accurate and dependable insights.

4. Conclusions

This project successfully achieved its main purpose of performing data preprocessing and exploratory data analysis on the Belgrade air pollution dataset. The dataset was found to be clean and well-structured, with no missing values, which made the preprocessing process smooth and reliable. Removing the city column reduced redundancy, while converting the time column to datetime allowed meaningful temporal analysis such as monthly distribution. These steps ensured that the data was properly prepared and suitable for further analysis and interpretation.

The exploratory analysis revealed clear patterns in pollutant behavior. Strong relationships were observed among nitrogen-based gases such as NO, NO_2, and NO_x, indicating that they likely originate from the same sources, mainly traffic and combustion processes. Particulate matter (PM_{10} and $PM_{2.5}$) also showed consistent behavior, suggesting shared emission sources and similar environmental influences. The presence of outliers across several pollutants was not random noise but rather realistic spikes caused by short-term events such as heavy traffic, industrial activity, or unfavorable weather conditions. This confirms that the dataset reflects genuine environmental variation.

Comparing manual analysis of the first 10 records with Python-based analysis of the complete 81-record dataset highlighted the importance of dataset size. While basic trends such as central tendency and general spread were visible in both, the smaller subset showed more fluctuation and less stability. The full dataset provided a more reliable and balanced view of pollution patterns, with smoother distributions and clearer relationships between variables. This comparison helped in understanding how limited data can sometimes exaggerate variability or hide true trends.

Feature scaling successfully normalized all numerical variables, bringing them to a common scale. Although scaling did not change the underlying patterns or relationships in the data, it prepared the dataset for future machine learning applications where equal feature contribution is essential. Overall, the analysis provided valuable insight into air quality conditions in Belgrade during the winter period, showing that emission sources have a stronger impact than meteorological factors. The project demonstrates how systematic preprocessing and EDA can transform raw environmental data into meaningful information for decision-making and further research.

5. Recommendations

The analysis can be extended by applying more advanced data mining techniques such as clustering (for example, K-means) on the scaled dataset to identify different pollution patterns. This would help group days with similar air quality conditions and improve understanding of pollution behaviour.

Future work should include collecting data for a longer time period, such as multiple seasons or years, to observe long-term trends and seasonal changes more clearly. A larger dataset would also improve the reliability of any further analysis or predictions.

For deeper insight, the dataset could be combined with additional information such as traffic volume, industrial activity, or weather forecasts. This would allow better identification of pollution sources and improve decision-making for air quality control and environmental planning.

6. References

1. NSW Environment & Heritage, *Understanding Air Quality Data: Units and Conversion Factors*, Environment New South Wales, Sydney, Australia, Available: https://www.environment.nsw.gov.au/topics/air/understanding-air-quality-data/units-and-conversion-factors
2. Balkan Green Energy News, *World Bank: Main Sources of Air Pollution in Belgrade are Heating, Road Transport and Coal Power Plants*, Balkan Green Energy News, 2021, Belgrade, Serbia, Available: https://balkangreenenergynews.com/world-bank-main-sources-of-air-pollution-in-belgrade-are-heating-road-transport-coal-power-plants/
3. DataCamp, *Python Boxplots Tutorial*, DataCamp Inc., 2023, New York, USA, Available: https://www.datacamp.com/tutorial/python-boxplots

7. Appendices

A. Complete Python Code

```
import pandas as pd
df=pd.read_csv("azzsdata2022.txt",sep="\t")

df.head()
```

```
df.info()
```

```
df.drop(columns=['city'], inplace=True)
df.head()
```

```
df.describe()
```

```
df['time'] = pd.to_datetime(df['time'], dayfirst=True)
df.head()
```

```
df.info()
```

```
import matplotlib.pyplot as plt

month_counts = df['time'].dt.to_period('M').value_counts().sort_index()
plt.figure(figsize=(10,6))
plt.bar(month_counts.index.astype(str), month_counts.values)
plt.title('Distribution of Records by Month', fontsize=16, fontweight='bold')
plt.xlabel('Month')
plt.ylabel('Number of Records')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

print("Month distribution:")
print(month_counts)
```

```
# Select only numeric columns
numeric_df = df.select_dtypes(include=['float64', 'int64'])
numeric_df.hist(figsize=(12,10),bins=30,edgecolor="black")
plt.subplots_adjust(hspace=0.7,wspace=0.4)
```

```
import seaborn as sns

plt.figure(figsize=(10,8))
sns.heatmap(numeric_df.corr(), annot=True, fmt=".2f", cmap="coolwarm")
plt.title("Correlation Heatmap of Air Pollution Variables (Belgrade)")
plt.show()
```

```
numeric_cols = list(df.select_dtypes(include=['float64', 'int64']).columns)

sns.pairplot(df[numeric_cols], diag_kind="kde")
plt.suptitle("Pairplot of Air Pollution and Weather Variables",
             fontsize=18, color="red", y=1.02)
plt.show()
```

```
fig, axes = plt.subplots(4, 3, figsize=(15, 12))  # 4 rows, 3 cols
axes = axes.flatten()

for i, col in enumerate(numeric_cols):
    axes[i].boxplot(df[col].dropna())
    axes[i].set_title(col)
    axes[i].set_ylabel('Values')

# Remove empty subplot if necessary
for j in range(i+1, len(axes)):
    fig.delaxes(axes[j])

plt.tight_layout()
plt.show()
```

```
from sklearn.preprocessing import StandardScaler
numeric_columns = df.select_dtypes(include=['int64', 'float64']).columns

scaler = StandardScaler()
df_scaled = df.copy()
df_scaled[numeric_columns] = scaler.fit_transform(df[numeric_columns])
```

```
fig, axes = plt.subplots(4, 3, figsize=(15, 12))  # 4 rows, 3 columns
axes = axes.flatten()

for i, col in enumerate(numeric_cols):
    axes[i].boxplot(df_scaled[col].dropna())
    axes[i].set_title(col)
    axes[i].set_ylabel('Scaled Values')

# Remove empty subplots if any
for j in range(i+1, len(axes)):
    fig.delaxes(axes[j])

plt.tight_layout()
plt.show()
```

B. Manual Data Description & EDA